

ROLE OF SANSKRIT IN NATURAL LANGUAGE PROCESSING Natural Language Processing (NLP) is a branch of linguistics mainly concerned with the processing of natural language data using computers and programming techniques. NLP uses ideas from linguistics, computer science, and artificial intelligence to make natural languages understandable to computers so that computers can carry out actions based on the instructions given in a natural language. NLP can be viewed as two sub-sets of ideas; Natural Language Generation (NLG) is the process of generating text from a meaning representation using a logic implemented through computer programs. On the contrary, Natural Language Understanding (NLU) is the process of distilling the meaning from natural language data using computer programs. Both are complementary and together form the basic building block for NLP. NLG is comparatively easier to handle than NLU, especially when there are rule-based constructs, lexicon, and syntax available for word generation. NLU, on the other hand, involves handling ambiguities.

The underlying logic for NLP is to develop a robust methodology for information processing, coding the information, disintegrating them into components, Identifying appropriate meanings for the components, and ensuring that the meanings are related unambiguously, and the meaning logically flows through the sentence. This is easier said than done and there are multiple challenges that one will face in the process. Let us take an example of a sentence in Sanskrit to understand some of these issues in the sentence. बालः वृक्षस्य फल खादति (balaḥ vrkṣasya phalaṁ khadati) meaning, the boy eats the fruit of the tree, there are some components and information encoded. One of the achievements of the early linguists was that they understood this process of information encoding and they assigned those meanings to the components. The rule-based derivational aspect of the language and the principles of karaka and vibhakti will enable us to distill the components and the encoded information in the sentence as given below:

बाल	सुँ	वृक्ष	स्य	फल	अम्	खाद्	ति
bāla	Sū	Vṛkṣa	Sya	phala	am	khād	Ti
boy	-	Tree	of	fruit	karma	To eat	Kartṛ
Noun base	1st case suffix	Noun base	6th case suffix	Noun base	2nd case suffix	Verb base	Present tense suffix

Once it is understood, the next step is to see how these meanings are related and then how the information flows through the sentence. In the above example, these are merely pieces of information in the sentence. They don't readily convey the information the speaker wants to convey. It is important to know how they are related. The structure of the sentence conveys that also. The concepts encoded in the components of these words have expectancy for other meanings by nature, and hence they find their connection with other meanings. This expectancy is called *akankṣa*. Indian linguists had understood that a *kriya* (action) always has expectancy for *karaka*s (participants). So, whenever there is a '*kriya*' indicated by a *dhatu*, it is looking for a *karaka* to relate. When a *karaka* is found marked by the second case, since it also has expectancy, they both find the relation. Hence we come to know that it is the fruit that the boy is eating not the tree because the tree is not related to eating directly. Such groups of semantically

connected concepts help in understanding the actual meaning of a sentence. This is the way the information in a sentence is processed in a natural language. This understanding is very essential for the processing of information by computer. It is difficult for a computer to comprehend natural languages like humans as there are many ambiguities in natural languages. Humans can easily resolve these ambiguities as they are aware of the context whereas a computer-based program cannot. This inability to comprehend natural language results in a huge limitation in achieving the goal of NLP based information in a sentence. Indian linguists have described fourteen determiners to fix the meaning of a word in case of activities such as machine translation. Humans can remove the ambiguity by the context. 'Payas' is the common word for both water and milk in Sanskrit. When we hear the sentence नद्या पयः प्रवहति (nadyam payah pravahati), meaning 'payas' is flowing in the river, we immediately resolve the ambiguity because we know for the fact that it is water that flows in a river, not milk. Indian linguists have described fourteen such determiners to fix the meaning of a word in case of multiple meanings. These are useful features for word sense disambiguation routines used in NLP.

Studies show that certain inherent advantages in the structure of the Sanskrit language and the grammar makes it an attractive candidate for NLP and artificial intelligence-related work. The Sanskrit language has certain inherent features such as being derivational and inflectional. With the availability of a full-fledged generative grammar for Sanskrit in the form of Astádhyañi, it lends itself as a potential candidate for use in NLP. Despite these inherent advantages, there is still a lot of work to accomplish to use Sanskrit in NLP. In recent years several researchers across the world have worked in the area of Sanskrit computational linguistics and have developed several components that can aid NLP using Sanskrit. This includes a variety of word generators (including compound words), morphological analysers, parsers, and creating lexical resources.